Discrete Structure (CST-002) - Unit 4

Unit 4 - Algebraic Structures (Groups & Rings) - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: algebraic structures - semigroups, monoids, groups, subgroups, cyclic groups, cosets and Lagrange's theorem, then rings, integral domains and fields.

Key ideas: A set with an operation is a **semigroup** if the operation is closed and associative; add an **identity** element and it is a **monoid**; add an **inverse** for every element and it is a **group**. If the operation also commutes, it is **abelian**. A **subgroup** is a subset that is itself a group (check closure and inverses). A group generated by one element is **cyclic**. **Lagrange's theorem:** the order of a subgroup divides the order of the group - used constantly. Moving up: a **ring** has two operations (like + and ×) where + is an abelian group and × distributes over +; an **integral domain** adds commutativity, identity and no zero-divisors; a **field** is a ring where every non-zero element has a multiplicative inverse.

Exam pointers: "prove (G, *) is a group" = tick the four axioms one by one with named steps. Cosets + Lagrange and the ring/domain/field chain are favourite theory questions - see the simple notes written for Integral Domain & Field below in this page.

Extra note - Integral Domain & Field - made simple:

First, a ring: a set of numbers where **both + and x** work nicely (like integers: 1+2=3, 2x3=6).

Zero divisor: two non-zero numbers whose product is **0**. This never happens with normal integers. But in Z6 (clock numbers 0-5): `2 x 3 = 6 = 0` - so 2 and 3 are zero divisors.

Integral domain: a ring with **no** zero divisors. Example: Z (all integers) - here a x b = 0 only when a or b is itself 0.

Field: a ring where every non-zero number has its **inverse** in the same set - meaning you can divide.

> *Small example: in Z5 (numbers 0-4), check: 2 x 3 = 6 = 1, so the inverse of 2 is 3. And 4 x 4 = 16 = 1, so 4 is its own inverse. Every element has an inverse - **Z5 is a field**. Z (integers) is not a field, because the inverse of 2 is 1/2, which is not an integer.*

Rule to remember: every field is an integral domain (if inverses exist, zero divisors cannot). Every finite integral domain is always a field.